

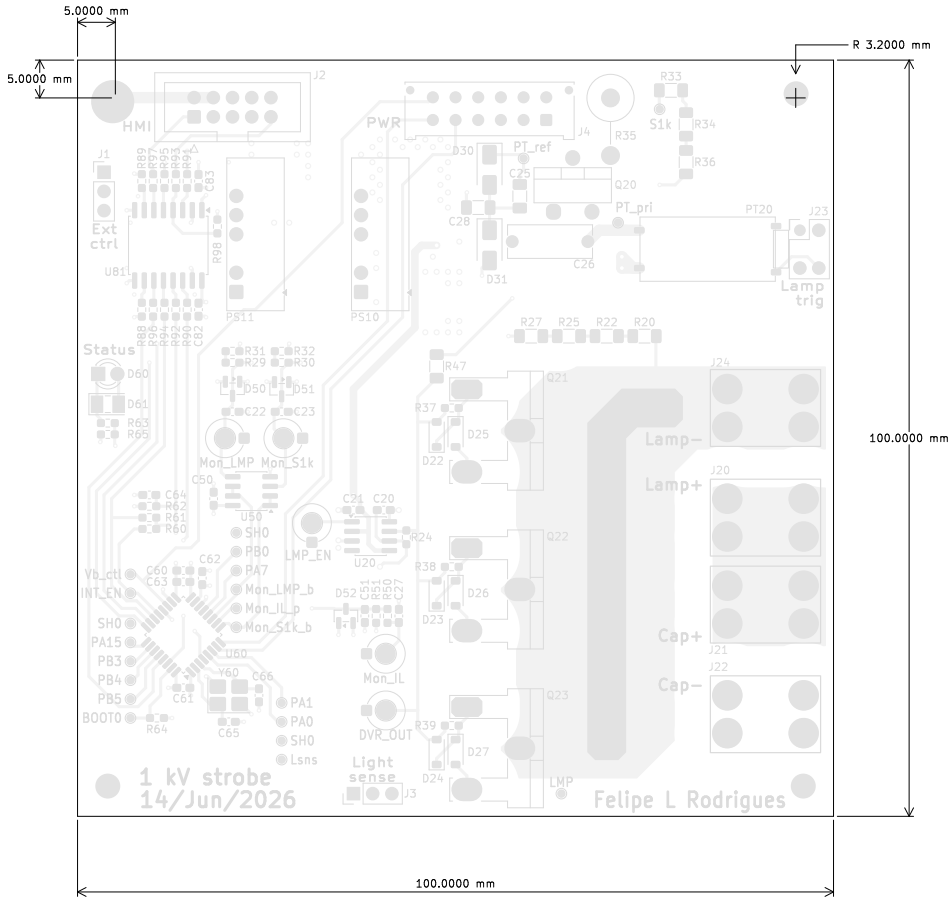
Main board Fabrication Document

Top Fabrication (Scale 1:1)

Layer Stack Legend

Material	Layer	Thickness	Dielectric	Type	Gerber
F.Paste				Paste Mask	
F.Silkscreen				Legend	GBR
F.Mask		0.02mm	Solder Resist	Solder Mask	GBR
Copper	F.Cu (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	In1.Cu (GND)	0.035mm (1.00oz)		Plane	GBR
Core		1.828mm	FR4	Dielectric	
Copper	In2.Cu (Sig, PWR)	0.035mm (1.00oz)		Plane	GBR
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	B.Cu (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
B.Mask		0.02mm	Solder Resist	Solder Mask	GBR
B.Silkscreen			Direct Printing	Legend	GBR
B.Paste				Paste Mask	

Total thickness: 2.278mm
Note: external layer thicknesses are specified after plating



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- FABRICATE PER IPC-6012A CLASS 2.
- OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts,GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- SURFACE FINISH: HAL SNPB
- SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
- SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- PCB MATERIAL REQUIREMENTS:
 - FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
 - Tg 170 C OR EQUIVALENT.
 - EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CHURRASCOD ELECTRONICS.
- DESIGN GEOMETRY MINIMUM FEATURE SIZES:

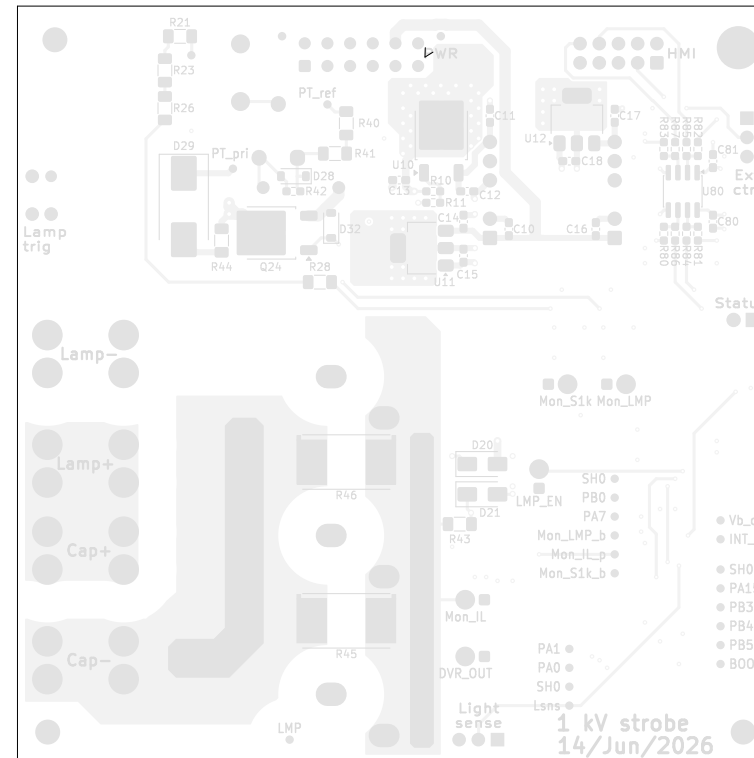
BOARD SIZE	100.000 x 100.000 mm
BOARD THICKNESS	2.278 mm
TRACE WIDTH	0.200 mm
TRACE TO TRACE	-0.000 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	1.000 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.300 mm
COPPER TO EDGE	0.500 mm
HOLE TO HOLE	0.250 mm



Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: ab924f9
	Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Top Fabrication (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13
Sheet Path:		Reviewer:	Size: A3	Revision: 1.0.0+ (Unreleased)
			Sheet: 1 of 10	

Main board Fabrication Document

Bottom Fabrication (Scale 1:1)



1 kV strobe
04/Jul/25
Felipe L Rodrigues



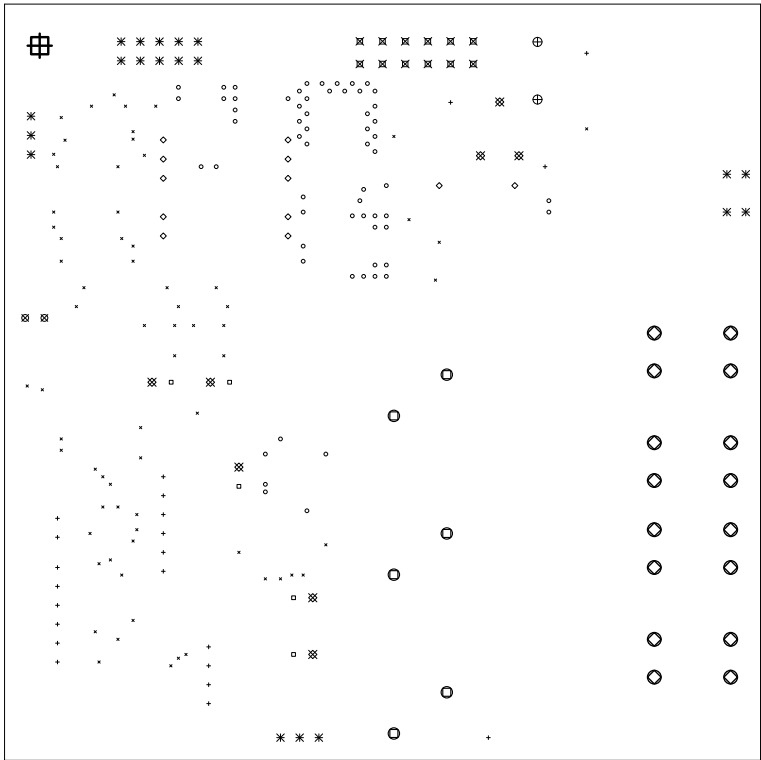
Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: ab924f9
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 2 of 10

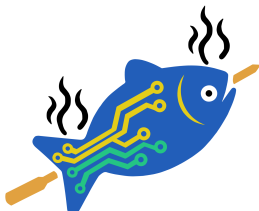
Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	69	0,30mm (11,81mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
○	62	0,50mm (19,69mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
+	22	0,50mm (19,69mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
□	5	0,70mm (27,56mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
◇	12	0,80mm (31,50mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊗	2	0,90mm (35,43mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
※	20	1,00mm (39,37mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊠	12	1,02mm (40,16mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊞	8	1,10mm (43,31mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊕	2	1,20mm (47,24mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
○	6	1,50mm (59,06mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
○	16	1,80mm (70,87mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊕	1	3,20mm (125,98mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
	Total 237					





ChurrasCod
Electronics

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Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 3 of 10

Main board Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	1,02mm (40,16mils)	NPTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Mechanical
○	1	1,35mm (53,15mils)	NPTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Mechanical
+	3	3,20mm (125,98mils)	NPTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Mechanical
	Total 6					

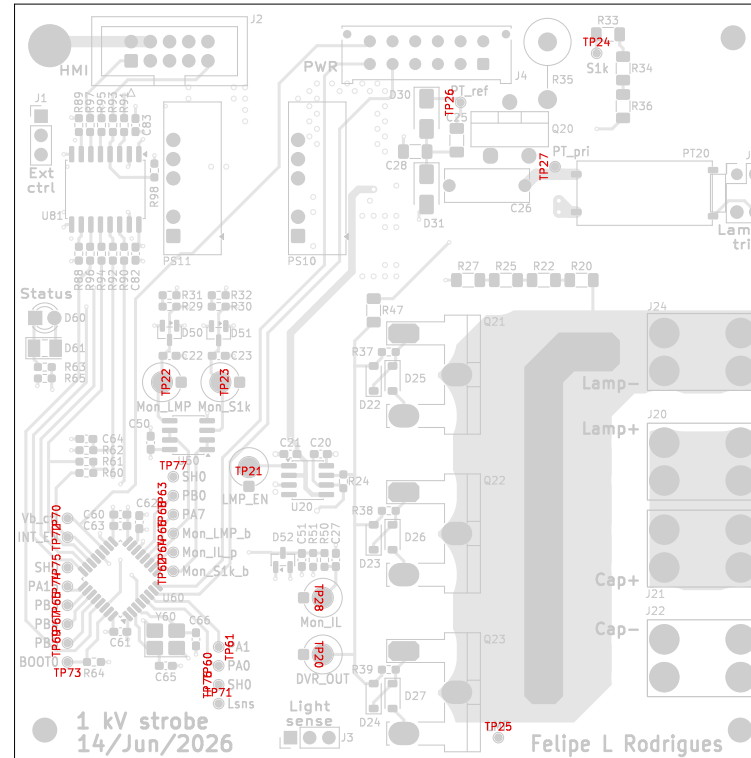
Drill Drawing L1 - L4 (Scale 1:1)

ChurrasCod Electronics

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	Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13
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Main board Fabrication Document

Top Test Points (Scale 1:1)



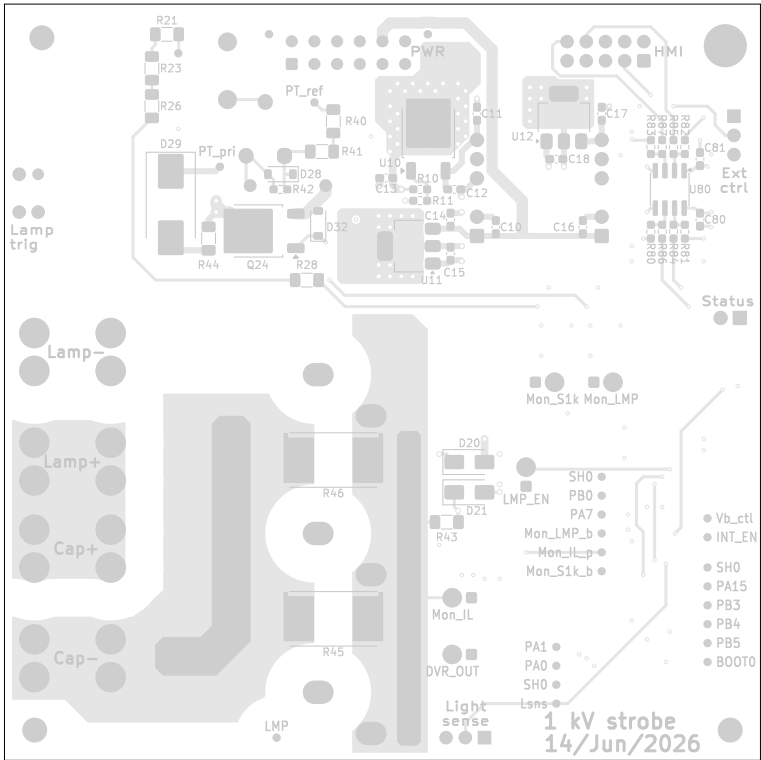
Ref.	Net	X [mm]	Y [mm]
TP20	SH0	44.50	19.00
TP21	SH0	36.00	42.50
TP22	SH0	25.77	55.00
TP23	Mon_S1k	33.50	55.00
TP24	S1k	82.00	98.50
TP25	LMP	69.00	8.00
TP26	TP_REF	64.00	92.00
TP27	TP_PRI	76.50	83.50
TP28	Mon_IL	44.50	26.50
TP60	Net-(U60-PA0)	32.00	17.50
TP61	Net-(U60-PA1)	32.00	20.00
TP62	Mon_S1k_buf	26.00	30.00
TP63	Net-(U60-PB0)	26.00	40.00
TP64	Mon_IL_prot	26.00	32.50
TP65	Net-(U60-PB3)	12.00	25.50
TP66	Mon_LMP_buf	26.00	35.00
TP67	Net-(U60-PB4)	12.00	23.00
TP68	Net-(U60-PA7)	26.00	37.50
TP69	Net-(U60-PB5)	12.00	20.50
TP70	Vbus_ctrl	12.00	37.00
TP71	Light_sns	32.00	12.50
TP72	INT_EN	12.00	34.50
TP73	BOOT0	12.00	18.00
TP74	Net-(U60-PA15)	12.00	28.00
TP75	SH0	12.00	30.50
TP76	SH0	32.00	15.00
TP77	SH0	26.00	42.50



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Main board Fabrication Document

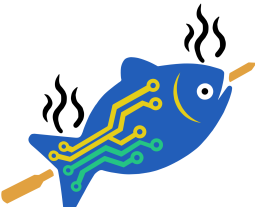
Bottom Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
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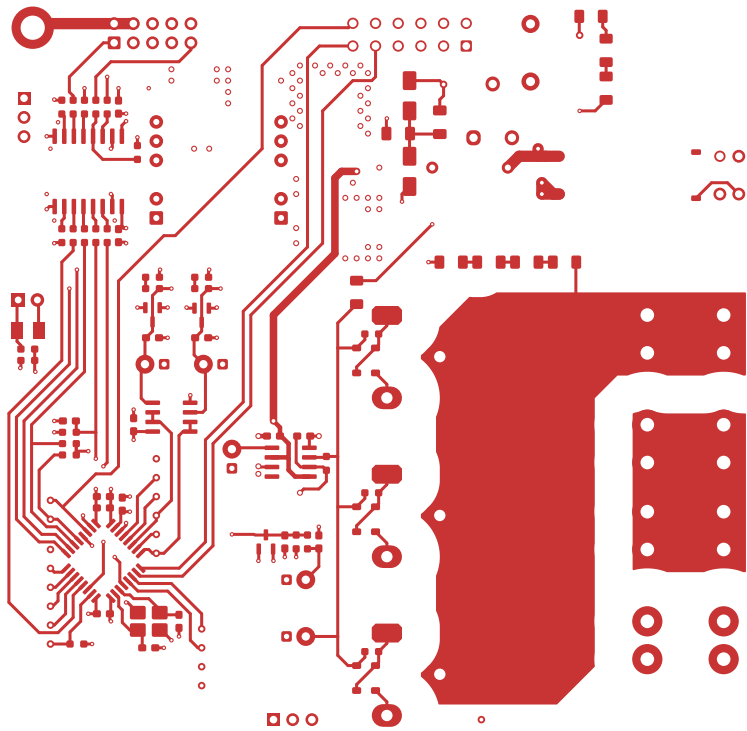


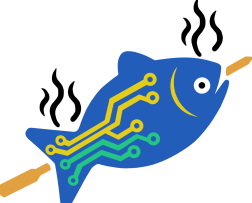
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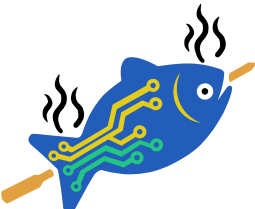
Main board Fabrication Document

F.Cu (Sig, PWR) (Scale 1:1)



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	Sheet Title: F.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 7 of 10

1	2	3	4	5	6	7	8
Main board Fabrication Document							In1.Cu (GND) (Scale 1:1)
A							
B							
C							
D							
E							
F	1	2	3	4	5	6	7

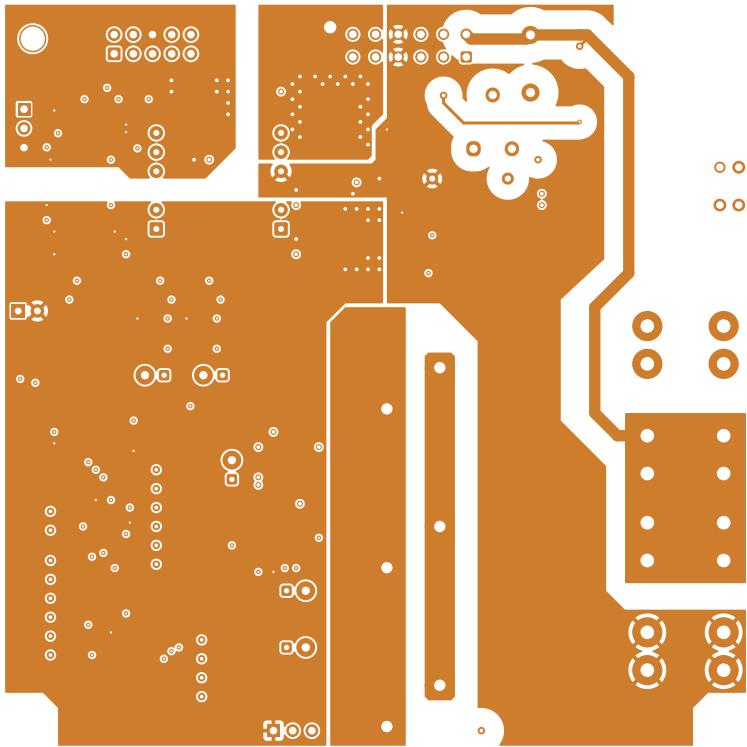


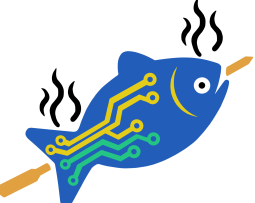
ChurrasCod Electronics

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	Board Name: Main board		Project Name: 1 kV strobe		
	Sheet Title: In1.Cu (GND) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
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Main board Fabrication Document

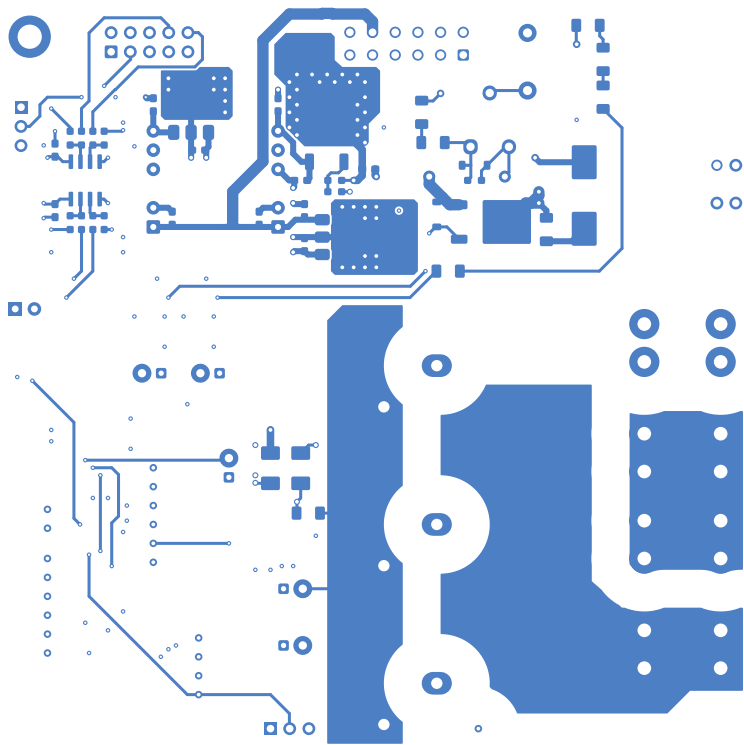
In2.Cu (Sig, PWR) (Scale 1:1)

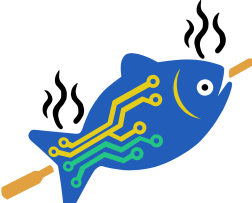


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	Sheet Title: In2.Cu (Sig, PWR) (Scale 1:1)		File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:			Reviewer:	Size: A3	Sheet: 9 of 10

Main board Fabrication Document

B.Cu (Sig, PWR) (Scale 1:1)



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	Sheet Title: B.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb		Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:			Reviewer:	Size: A3	Sheet: 10 of 10